

EDUCATION

- **Huazhong University of Science and Technology**

Bachelor of Engineering in Computer Science and Technology

GPA: 4.70/5.00, Rank: 1/30

Honors: Outstanding Undergraduates, China National Scholarship (Top 0.2% nationwide)

Wuhan, China

Sept. 2023 – Jun. 2027

EXPERIENCE

- **National University of Singapore**

A+ Evaluation, NUS SOC Summer Workshop

Singapore

July 2025 – Aug. 2025

- Collected, cleaned, and preprocessed a multi-class image dataset for scenarios including smoking, fighting, falling, and littering, and trained/fine-tuned models using the **YOLOv7 framework** to optimize detection accuracy and speed for specific scenarios.
- Implemented multi-model object detection and the **SORT tracking algorithm** to ensure robust target tracking in complex scenes; developed a **sliding-window algorithm** for video streams to enable effective recognition of continuous behaviors.
- Designed and implemented an **end-edge-cloud architecture**, where the robot transmitted data efficiently to the server via **MQTT** for inference, and fed results back to both the monitoring client and the robot.
- Built the monitoring client using **Dear PyGui**, covering the full pipeline from video input and model inference to result visualization.

PROJECTS

- **RISC-V Proxy Operating System Kernel**

Jan. 2026 – Mar. 2026

- A **multi-core concurrency-safe** proxy operating system kernel developed for the **Spike** target simulator based on the **RISC-V architecture**.
- Implemented **Sv39 virtual memory** and **copy-on-write (COW)**, reducing overhead while preserving semantic correctness through page table permission control, page-fault-triggered copying, and page reference counting.
- Built process management and scheduling modules, including **fork**, **yield**, **wait**, and **round-robin time slicing**, and introduced synchronization mechanisms such as **semaphores** and **spinlocks**.
- Designed a **VFS (Virtual File System)** abstraction layer supporting basic file operations (**open**, **read**, **write**, **close**) as well as files, directories, links, and relative paths.
- Developed an interactive **Shell** on top of the kernel, supporting command history, environment variables, and basic built-in commands; designed a **kernel-assisted pipe mechanism** for inter-process communication and implemented **dynamic scheduling of background tasks on multiple cores**.

- **RMDB Relational Database Management System**

May 2025 – Aug. 2025

- Implemented SQL lexical and syntax parsing based on **Flex/Bison**, constructed the AST, and completed semantic analysis including column binding, type checking, and conditional expression validation.
- Built a rule-based query optimizer and a **Volcano-model** execution engine, supporting operators such as **predicate pushdown**, **projection pushdown**, **index selection**, and **join plan generation**.
- Designed a **page-based storage architecture**, **buffer pool**, and **B+ tree composite indexes**, supporting equality queries, range scans, and leftmost-prefix matching.
- Implemented transaction rollback and **MVCC-based** snapshot visibility control, and built a logging and recovery framework to support database crash recovery.

PROGRAMMING SKILLS

- **Languages:** C/C++, Python, Golang, Java, Swift, SQL
- **Skills:** Docker, Gin Web, Graph Transformer, Pytorch, Git, CMake, Makefile, Neovim